

Data Warehousing and Mining

Lecture 1

- Course syllabus
- Overview of data warehousing and mining

Lecture slides modified from:

- Jiawei Han (http://www-sal.cs.uiuc.edu/~hanj/DM_Book.html)
- Vipin Kumar (<http://www-users.cs.umn.edu/~kumar/csci5980/index.html>)
- Joshep Fong (<http://www.cs.cityu.edu.hk/~jfong/course/cs5483/>)
- Slobodan Vucetic (<http://www.ist.temple.edu/~vucetic/cis526fall2004.htm>)

Course Syllabus

Textbook:

- 1- J. Han, M. Kamber, Data Mining: Concepts and Techniques, 2006, second edition.
- 2- Joseph Fong: Information Systems Reengineering and Integration, 2006, ISBN 978-1-84628-382-6, Second edition.

Topics:

- Overview of data warehousing and mining
- Data warehouse and OLAP technology
- Data preprocessing
- Mining association rules
- Classification and prediction
- Cluster analysis
- Mining complex types of data

Grading:

- (10%) Homework Assignments and Quizzes
- (30%) Project
- (60%) Final Exam.

Course Syllabus

Late Policy and Academic Honesty:

The projects and homework assignments are due in class, on the specified due date. NO LATE SUBMISSIONS will be accepted. For fairness, this policy will be strictly enforced.

Academic honesty is taken seriously. You must write up your own solutions and code. For homework problems or projects you are allowed to discuss the problems or assignments verbally with other class members. You **MUST** acknowledge the people with whom you discussed your work. Any other sources (e.g. Internet, research papers, books) used for solutions and code **MUST** also be acknowledged. In case of doubt **PLEASE** contact the instructor.

Disability Disclosure Statement

Any student who has a need for accommodation based on the impact of a disability should contact me privately to discuss the specific situation as soon as possible. Contact Disability Resources and Services at 215-204-1280 in 100 Ritter Annex to coordinate reasonable accommodations for students with documented disabilities.

Motivation:

“Necessity is the Mother of Invention”

- Data explosion problem
 - Automated data collection tools and mature database technology lead to tremendous amounts of data stored in databases, data warehouses and other information repositories
- We are drowning in data, but starving for knowledge!
- Solution: Data warehousing and data mining
 - Data warehousing and on-line analytical processing
 - Extraction of interesting knowledge (rules, regularities, patterns, constraints) from data in large databases

Why Mine Data? Commercial Viewpoint

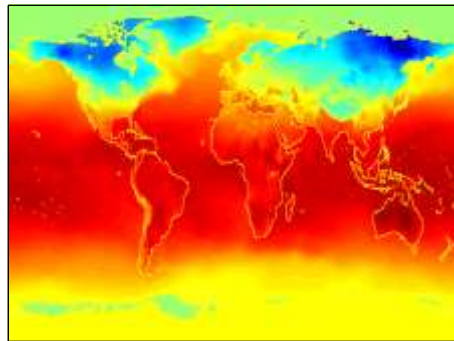
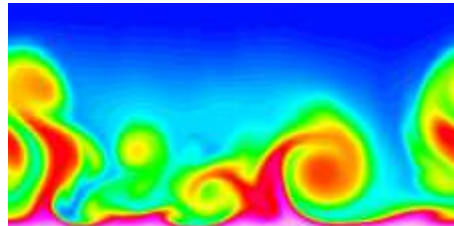
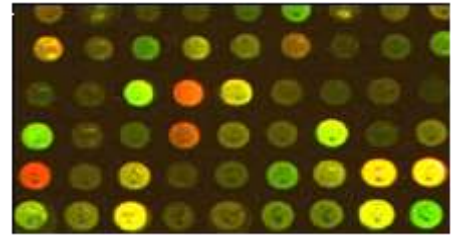
- Lots of data is being collected and warehoused
 - Web data, e-commerce
 - purchases at department/grocery stores
 - Bank/Credit Card transactions



- Computers have become cheaper and more powerful
- Competitive Pressure is Strong
 - Provide better, customized services for an *edge* (e.g. in Customer Relationship Management)

Why Mine Data? Scientific Viewpoint

- Data collected and stored at enormous speeds (GB/hour)
 - remote sensors on a satellite
 - telescopes scanning the skies
 - microarrays generating gene expression data
 - scientific simulations generating terabytes of data
- Traditional techniques infeasible for raw data
- Data mining may help scientists
 - in classifying and segmenting data
 - in Hypothesis Formation





What Is Data Mining?

- Data mining (knowledge discovery in databases):
 - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) information or patterns from data in large databases
- Alternative names and their “inside stories”:
 - Data mining: a misnomer?
 - Knowledge discovery(mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, business intelligence, etc.



Examples: What is (not) Data Mining?

● What is not Data Mining?

- Look up phone number in phone directory
- Query a Web search engine for information about “Amazon”

● What is Data Mining?

- Certain names are more prevalent in certain US locations (O’Brien, O’Rourke, O’Reilly... in Boston area)
- Group together similar documents returned by search engine according to their context (e.g. Amazon rainforest, Amazon.com,)

Data Mining: Classification Schemes

- Decisions in data mining
 - Kinds of databases to be mined
 - Kinds of knowledge to be discovered
 - Kinds of techniques utilized
 - Kinds of applications adapted
- Data mining tasks
 - Descriptive data mining
 - Predictive data mining

Decisions in Data Mining

- **Databases to be mined**
 - Relational, transactional, object-oriented, object-relational, active, spatial, time-series, text, multi-media, heterogeneous, legacy, WWW, etc.
- **Knowledge to be mined**
 - Characterization, discrimination, association, classification, clustering, trend, deviation and outlier analysis, etc.
 - Multiple/integrated functions and mining at multiple levels
- **Techniques utilized**
 - Database-oriented, data warehouse (OLAP), machine learning, statistics, visualization, neural network, etc.
- **Applications adapted**
 - Retail, telecommunication, banking, fraud analysis, DNA mining, stock market analysis, Web mining, Weblog analysis, etc.

Data Mining Tasks

- Prediction Tasks
 - Use some variables to predict unknown or future values of other variables
- Description Tasks
 - Find human-interpretable patterns that describe the data.

Common data mining tasks

- Classification [Predictive]
- Clustering [Descriptive]
- Association Rule Discovery [Descriptive]
- Sequential Pattern Discovery [Descriptive]
- Regression [Predictive]
- Deviation Detection [Predictive]

Classification: Definition

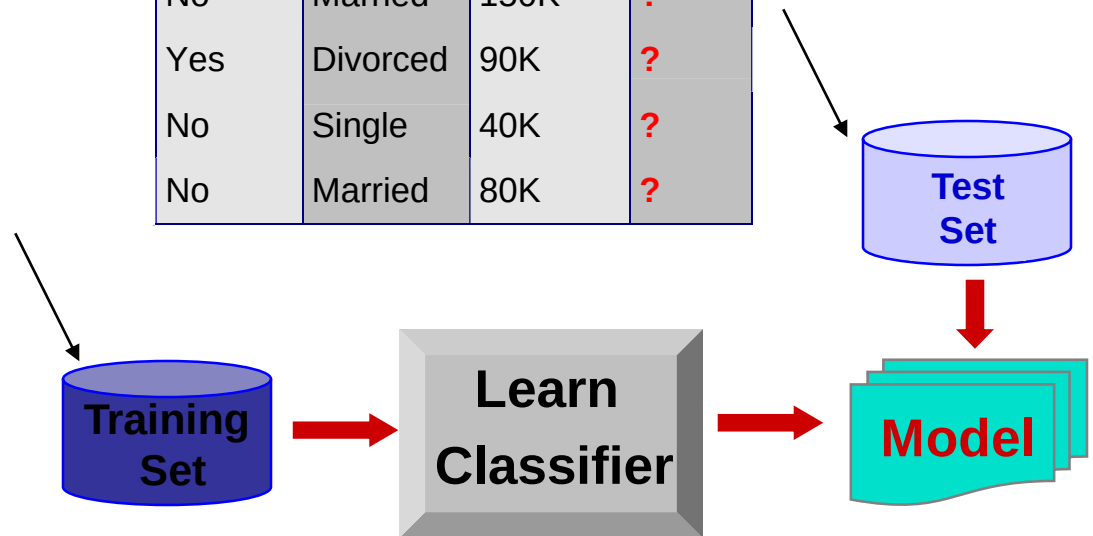
- Given a collection of records (*training set*)
 - Each record contains a set of *attributes*, one of the attributes is the *class*.
- Find a *model* for class attribute as a function of the values of other attributes.
- Goal: previously unseen records should be assigned a class as accurately as possible.
 - A *test set* is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it.

Classification Example

categorical
categorical
continuous
class

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Refund	Marital Status	Taxable Income	Cheat
No	Single	75K	?
Yes	Married	50K	?
No	Married	150K	?
Yes	Divorced	90K	?
No	Single	40K	?
No	Married	80K	?



Classification: Application 1

- Direct Marketing
 - Goal: Reduce cost of mailing by *targeting* a set of consumers likely to buy a new cell-phone product.
 - Approach:
 - Use the data for a similar product introduced before.
 - We know which customers decided to buy and which decided otherwise. This *{buy, don't buy}* decision forms the *class attribute*.
 - Collect various demographic, lifestyle, and company-interaction related information about all such customers.
 - Type of business, where they stay, how much they earn, etc.
 - Use this information as input attributes to learn a classifier model.

Classification: Application 2

- Fraud Detection
 - Goal: Predict fraudulent cases in credit card transactions.
 - Approach:
 - Use credit card transactions and the information on its account-holder as attributes.
 - When does a customer buy, what does he buy, how often he pays on time, etc
 - Label past transactions as fraud or fair transactions. This forms the class attribute.
 - Learn a model for the class of the transactions.
 - Use this model to detect fraud by observing credit card transactions on an account.

Classification: Application 3

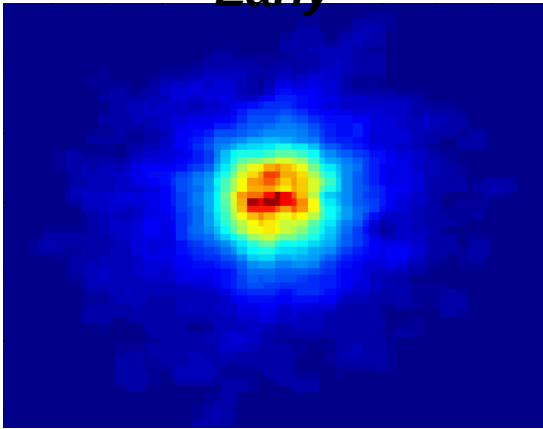
- Customer Attrition/Churn:
 - Goal: To predict whether a customer is likely to be lost to a competitor.
 - Approach:
 - Use detailed record of transactions with each of the past and present customers, to find attributes.
 - How often the customer calls, where he calls, what time-of-the day he calls most, his financial status, marital status, etc.
 - Label the customers as loyal or disloyal.
 - Find a model for loyalty.

Classification: Application 4

- Sky Survey Cataloging
 - Goal: To predict class (star or galaxy) of sky objects, especially visually faint ones, based on the telescopic survey images (from Palomar Observatory).
 - 3000 images with 23,040 x 23,040 pixels per image.
 - Approach:
 - Segment the image.
 - Measure image attributes (features) - 40 of them per object.
 - Model the class based on these features.
 - Success Story: Could find 16 new high red-shift quasars, some of the farthest objects that are difficult to find!

Classifying Galaxies

Early



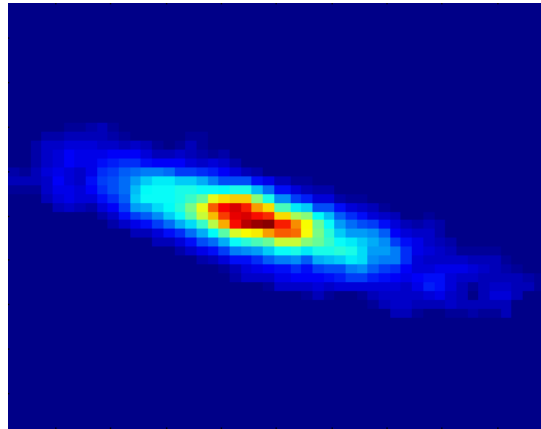
Class:

- Stages of Formation

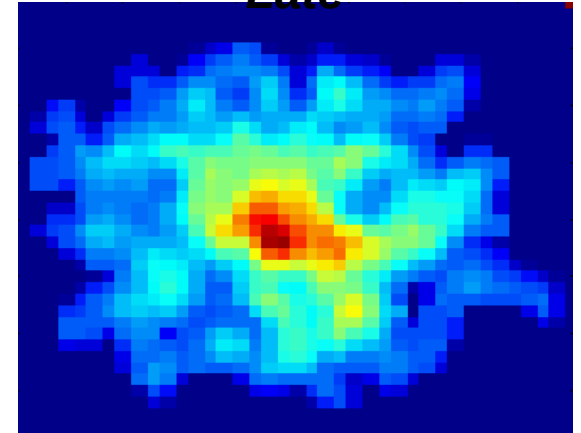
Attributes:

- Image features,
- Characteristics of light waves received, etc.

Intermediate



Late



Data Size:

- 72 million stars, 20 million galaxies
- Object Catalog: 9 GB
- Image Database: 150 GB

Clustering Definition

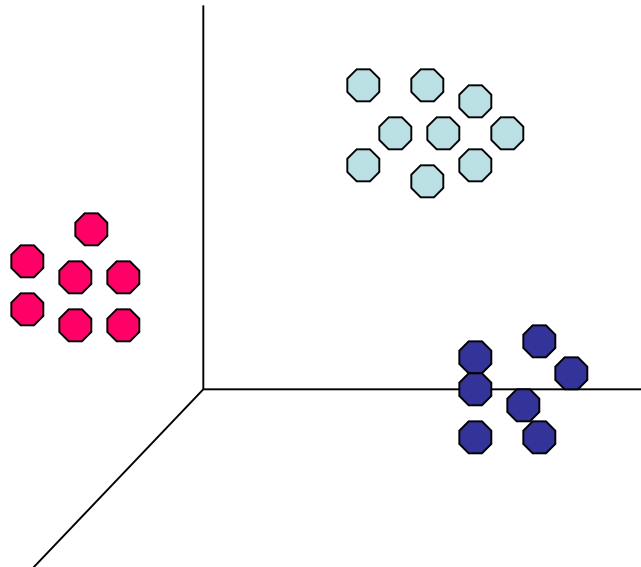
- Given a set of data points, each having a set of attributes, and a similarity measure among them, find clusters such that
 - Data points in one cluster are more similar to one another.
 - Data points in separate clusters are less similar to one another.
- Similarity Measures:
 - Euclidean Distance if attributes are continuous.
 - Other Problem-specific Measures.

Illustrating Clustering

☒ Euclidean Distance Based Clustering in 3-D space.

Intracuster distances
are minimized

Intercluster distances
are maximized



Clustering: Application 1

- Market Segmentation:
 - Goal: subdivide a market into distinct subsets of customers where any subset may conceivably be selected as a market target to be reached with a distinct marketing mix.
 - Approach:
 - Collect different attributes of customers based on their geographical and lifestyle related information.
 - Find clusters of similar customers.
 - Measure the clustering quality by observing buying patterns of customers in same cluster vs. those from different clusters.

Clustering: Application 2

- Document Clustering:
 - Goal: To find groups of documents that are similar to each other based on the important terms appearing in them.
 - Approach: To identify frequently occurring terms in each document. Form a similarity measure based on the frequencies of different terms. Use it to cluster.
 - Gain: Information Retrieval can utilize the clusters to relate a new document or search term to clustered documents.

Association Rule Discovery: Definition

- Given a set of records each of which contain some number of items from a given collection;
 - Produce dependency rules which will predict occurrence of an item based on occurrences of other items.

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

{Milk} --> {Coke}

{Diaper, Milk} --> {Beer}

Association Rule Discovery: Application 1

- Marketing and Sales Promotion:
 - Let the rule discovered be
 $\{Bagels, \dots\} \rightarrow \{Potato\ Chips\}$
 - Potato Chips as consequent => Can be used to determine what should be done to boost its sales.
 - Bagels in the antecedent => Can be used to see which products would be affected if the store discontinues selling bagels.
 - Bagels in antecedent and Potato chips in consequent => Can be used to see what products should be sold with Bagels to promote sale of Potato chips!

Association Rule Discovery: Application 2

- Supermarket shelf management.
 - Goal: To identify items that are bought together by sufficiently many customers.
 - Approach: Process the point-of-sale data collected with barcode scanners to find dependencies among items.
 - A classic rule --
 - If a customer buys diaper and milk, then he is very likely to buy beer:

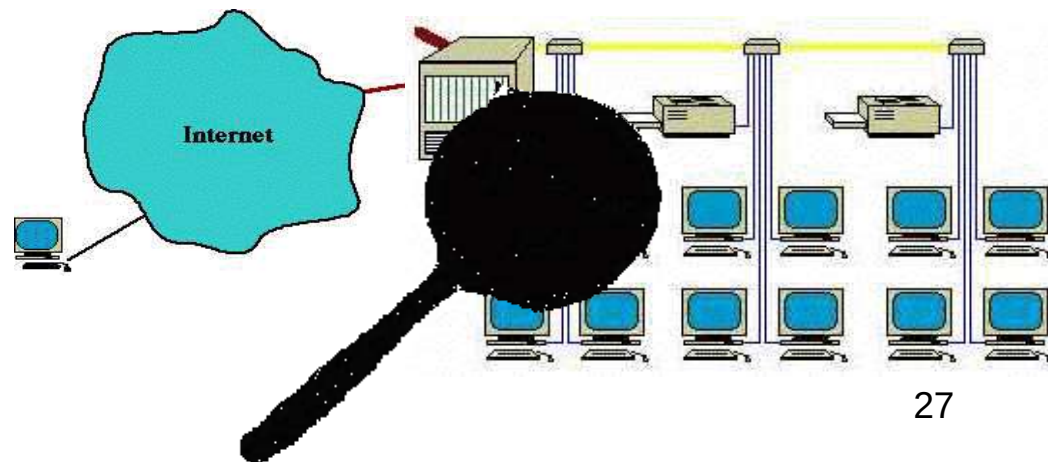
Diapers $\rightarrow$ Beer, support = 20%, confidence = 85%

Regression

- Predict a value of a given continuous valued variable based on the values of other variables, assuming a linear or nonlinear model of dependency.
- Greatly studied in statistics, neural network fields.
- Examples:
 - Predicting sales amounts of new product based on advertising expenditure.
 - Predicting wind velocities as a function of temperature, humidity, air pressure, etc.
 - Time series prediction of stock market indices.

Deviation/Anomaly Detection

- Detect significant deviations from normal behavior
- Applications:
 - Credit Card Fraud Detection
 - Network Intrusion Detection



Data Mining and Induction Principle

Induction vs Deduction

- Deductive reasoning is truth-preserving:
 1. All horses are mammals
 2. All mammals have lungs
 3. Therefore, all horses have lungs
- Induction reasoning adds information:
 1. All horses observed so far have lungs.
 2. Therefore, all horses have lungs.

The Problems with Induction

From true facts, we may induce false models.

Prototypical example:

- European swans are all white.
- Induce: "Swans are white" as a general rule.
- Discover Australia and black Swans...
- Problem: the set of examples is not random and representative

Another example: distinguish US tanks from Iraqi tanks

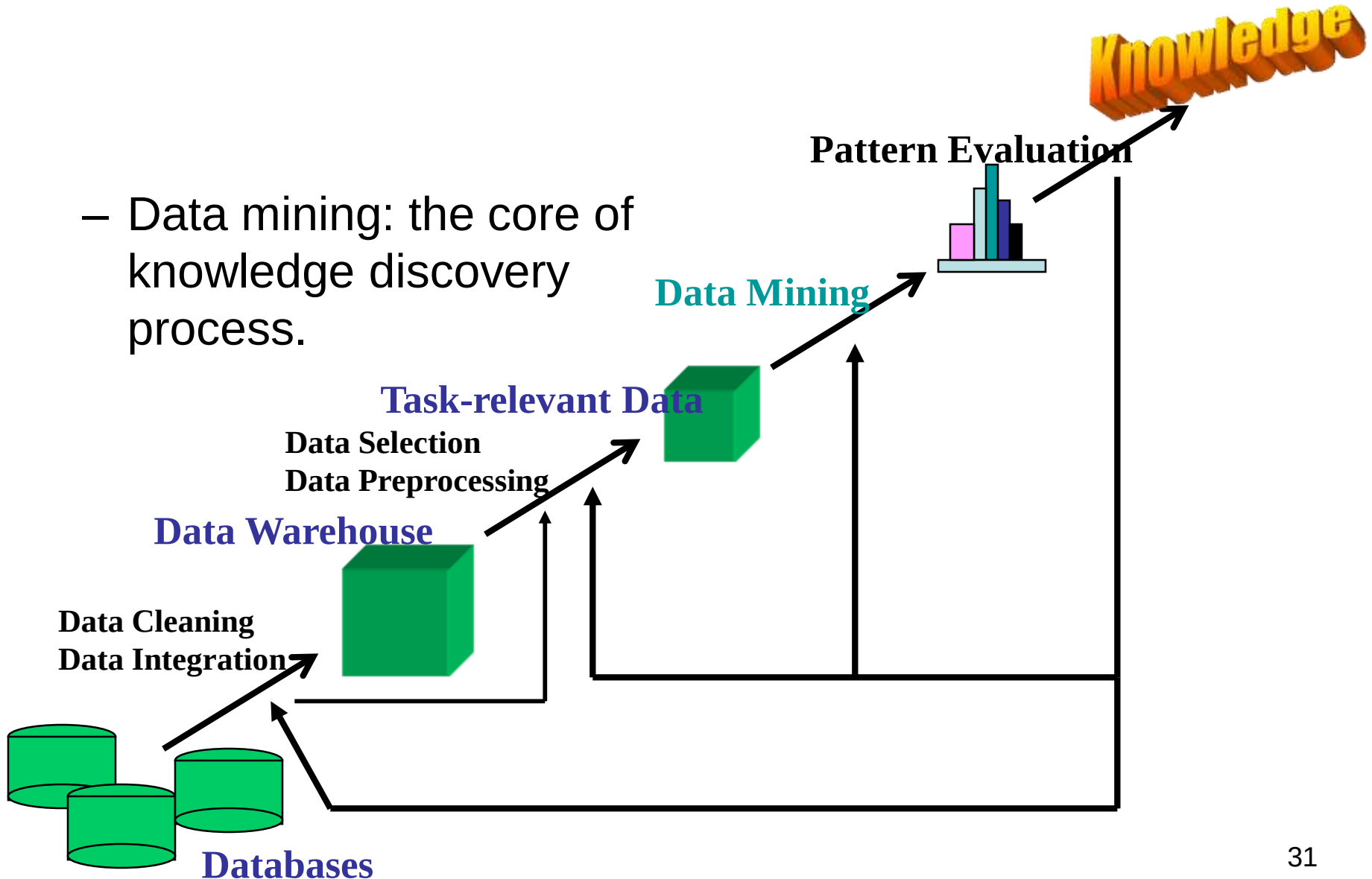
- Method: Database of pictures split in train set and test set; Classification model built on train set
- Result: Good predictive accuracy on test set; Bad score on independent pictures
- Why did it go wrong: other distinguishing features in the pictures (hangar versus desert)

Hypothesis-Based vs. Exploratory-Based

- The hypothesis-based method:
 - Formulate a hypothesis of interest.
 - Design an experiment that will yield data to test this hypothesis.
 - Accept or reject hypothesis depending on the outcome.
- Exploratory-based method:
 - Try to make sense of a bunch of data without an a priori hypothesis!
 - The only prevention against false results is significance:
 - ensure statistical significance (using train and test etc.)
 - ensure domain significance (i.e., make sure that the results make sense to a domain expert)

Data Mining: A KDD Process

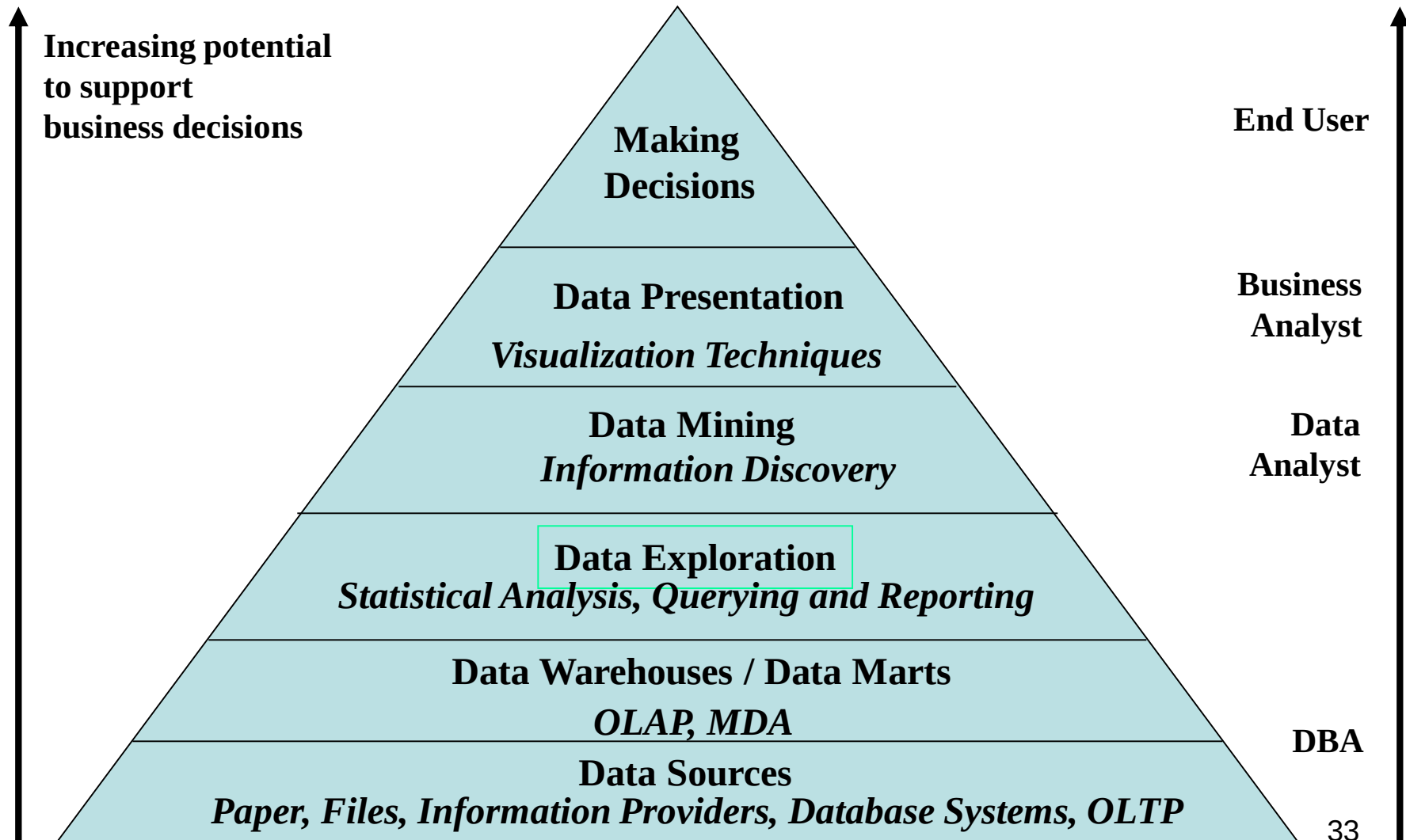
- Data mining: the core of knowledge discovery process.



Steps of a KDD Process

- Learning the application domain:
 - relevant prior knowledge and goals of application
- Creating a target data set: data selection
- **Data cleaning** and preprocessing: (may take 60% of effort!)
- **Data reduction and transformation:**
 - Find useful features, dimensionality/variable reduction, invariant representation.
- Choosing functions of data mining
 - summarization, classification, regression, association, clustering.
- Choosing the mining algorithm(s)
- **Data mining:** search for patterns of interest
- **Pattern evaluation and knowledge presentation**
 - visualization, transformation, removing redundant patterns, etc.
- Use of discovered knowledge

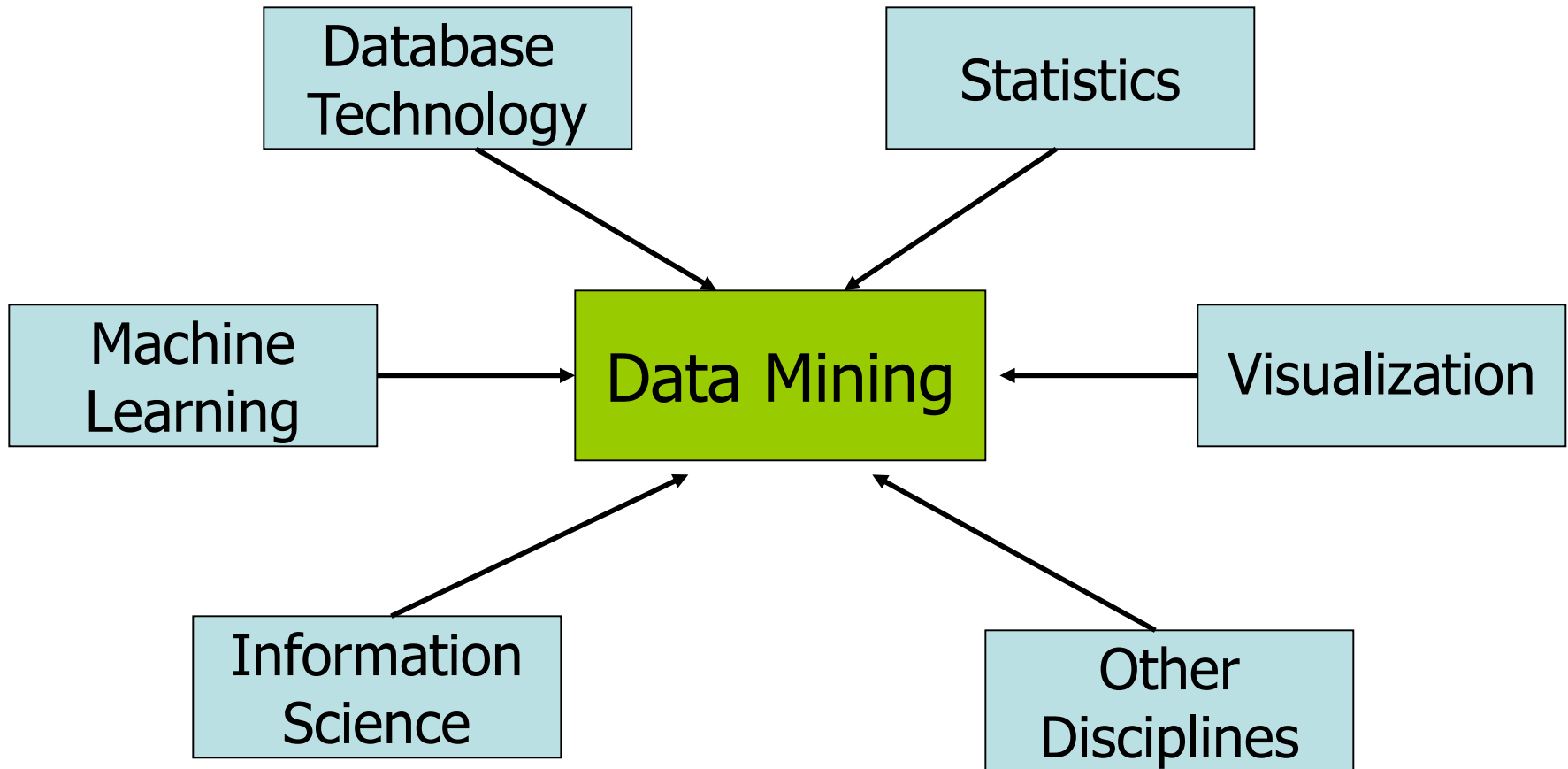
Data Mining and Business Intelligence



Data Mining: On What Kind of Data?

- Relational databases
- Data warehouses
- Transactional databases
- Advanced DB and information repositories
 - Object-oriented and object-relational databases
 - Spatial databases
 - Time-series data and temporal data
 - Text databases and multimedia databases
 - Heterogeneous and legacy databases
 - WWW

Data Mining: Confluence of Multiple Disciplines



Data Mining vs. Statistical Analysis

Statistical Analysis:

- Ill-suited for Nominal and Structured Data Types
- Completely data driven - incorporation of domain knowledge not possible
- Interpretation of results is difficult and daunting
- Requires expert user guidance

Data Mining:

- Large Data sets
- Efficiency of Algorithms is important
- Scalability of Algorithms is important
- Real World Data
- Lots of Missing Values
- Pre-existing data - not user generated
- Data not static - prone to updates
- Efficient methods for data retrieval available for use

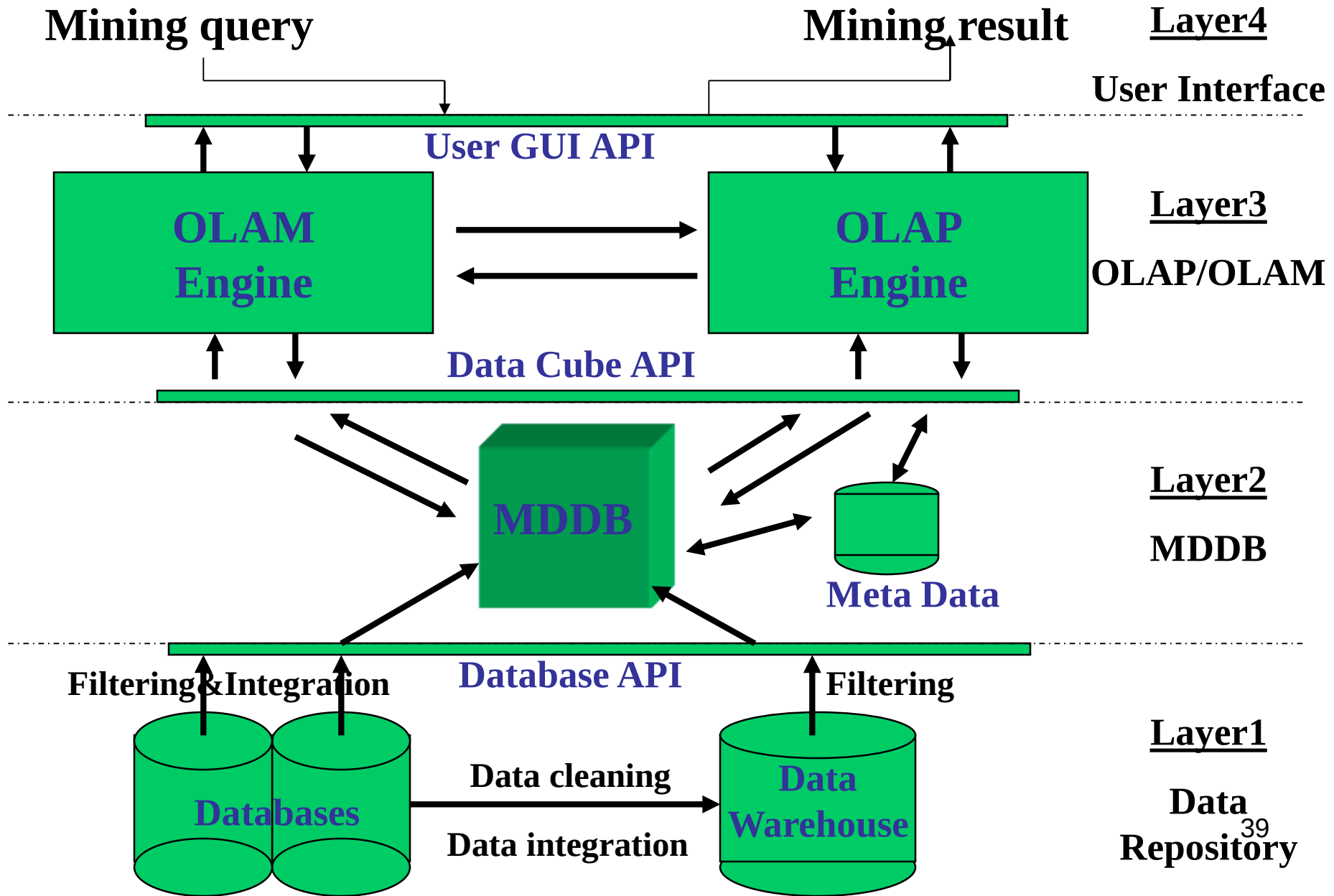
Data Mining vs. DBMS

- Example DBMS Reports
 - Last months sales for each service type
 - Sales per service grouped by customer sex or age bracket
 - List of customers who lapsed their policy
- Questions answered using Data Mining
 - What characteristics do customers that lapse their policy have in common and how do they differ from customers who renew their policy?
 - Which motor insurance policy holders would be potential customers for my House Content Insurance policy?

Data Mining and Data Warehousing

- Data Warehouse: a centralized data repository which can be queried for business benefit.
- Data Warehousing makes it possible to
 - extract archived operational data
 - overcome inconsistencies between different legacy data formats
 - integrate data throughout an enterprise, regardless of location, format, or communication requirements
 - incorporate additional or expert information
- OLAP: On-line Analytical Processing
- Multi-Dimensional Data Model (Data Cube)
- Operations:
 - Roll-up
 - Drill-down
 - Slice and dice
 - Rotate

An OLAM Architecture



DBMS, OLAP, and Data Mining

	DBMS	OLAP	Data Mining
Task	Extraction of detailed and summary data	Summaries, trends and forecasts	Knowledge discovery of hidden patterns and insights
Type of result	Information	Analysis	Insight and Prediction
Method	Deduction (Ask the question, verify with data)	Multidimensional data modeling, Aggregation, Statistics	Induction (Build the model, apply it to new data, get the result)
Example question	Who purchased mutual funds in the last 3 years?	What is the average income of mutual fund buyers by region by year?	Who will buy a mutual fund in the next 6 months and why?

Example of DBMS, OLAP and Data Mining: Weather Data

DBMS:

Day	outlook	temperature	humidity	windy	play
1	sunny	85	85	false	no
2	sunny	80	90	true	no
3	overcast	83	86	false	yes
4	rainy	70	96	false	yes
5	rainy	68	80	false	yes
6	rainy	65	70	true	no
7	overcast	64	65	true	yes
8	sunny	72	95	false	no
9	sunny	69	70	false	yes
10	rainy	75	80	false	yes
11	sunny	75	70	true	yes
12	overcast	72	90	true	yes
13	overcast	81	75	false	yes
14	rainy	71	91	true	no

Example of DBMS, OLAP and Data Mining: Weather Data

- By querying a DBMS containing the above table we may answer questions like:
- What was the temperature in the sunny days? {85, 80, 72, 69, 75}
- Which days the humidity was less than 75? {6, 7, 9, 11}
- Which days the temperature was greater than 70? {1, 2, 3, 8, 10, 11, 12, 13, 14}
- Which days the temperature was greater than 70 and the humidity was less than 75? The intersection of the above two: {11}

Example of DBMS, OLAP and Data Mining: Weather Data

OLAP:

- Using OLAP we can create a **Multidimensional Model** of our data (**Data Cube**).
- For example using the dimensions: **time**, **outlook** and **play** we can create the following model.

9 / 5	sunny	rainy	overcast
Week 1	0 / 2	2 / 1	2 / 0
Week 2	2 / 1	1 / 1	2 / 0

Example of DBMS, OLAP and Data Mining: Weather Data

Data Mining:

- Using the ID3 algorithm we can produce the following decision tree:
 - **outlook = sunny**
 - humidity = high: no
 - humidity = normal: yes
 - **outlook = overcast: yes**
 - **outlook = rainy**
 - windy = true: no
 - windy = false: yes

Major Issues in Data Warehousing and Mining

- Mining methodology and user interaction
 - Mining different kinds of knowledge in databases
 - Interactive mining of knowledge at multiple levels of abstraction
 - Incorporation of background knowledge
 - Data mining query languages and ad-hoc data mining
 - Expression and visualization of data mining results
 - Handling noise and incomplete data
 - Pattern evaluation: the interestingness problem
- Performance and scalability
 - Efficiency and scalability of data mining algorithms
 - Parallel, distributed and incremental mining methods

Major Issues in Data Warehousing and Mining

- Issues relating to the diversity of data types
 - Handling relational and complex types of data
 - Mining information from heterogeneous databases and global information systems (WWW)
- Issues related to applications and social impacts
 - Application of discovered knowledge
 - Domain-specific data mining tools
 - Intelligent query answering
 - Process control and decision making
 - Integration of the discovered knowledge with existing knowledge: A knowledge fusion problem
 - Protection of data security, integrity, and privacy